

Chapter III

TECHNICAL BACKGROUND

Backend Technologies Used

Visual Studio Code (VS Code). This is the code editor the researchers used to write, debug, and manage all of the backend's Java source code, SQL scripts, and configuration files.

Java. This is the programming language used to write the entire backend, including the logic that checks scanned answers, computes scores, and builds analytics reports.

Spring Boot. This is the Java framework used to build the backend as a set of RESTful API services that the mobile app and the web dashboard connect to.

Spring Web (Spring MVC). This is the part of Spring Boot used to expose the backend's features as REST API endpoints, such as downloading an assigned class to the mobile app or generating a report for the web dashboard.

Spring Security. This is the library used to check a logged-in user's role (teacher or principal) before allowing a request through, and to protect stored teacher passwords.

Spring JDBC (JdbcTemplate). This is the part of Spring used to connect the backend directly to the database and run the SQL statements that retrieve and save records such as students, assessments, and scanned answers.

Jakarta Bean Validation (Spring Boot Validation). This is the library used to check that data sent from the mobile app or the web dashboard is complete and correctly formatted before the backend processes it.

Spring Boot Mail. This is the library used to send account verification and registration emails to newly created teacher accounts.

Spring Boot DevTools. This is the tool used to automatically restart the backend server whenever the researchers saved a code change during development.

MySQL / MariaDB. This is the relational database management system used to permanently store all of the system's data, such as users, classes, students, assessments, scanned answers, and analytics results.

Apache XAMPP. This is the local server package used to run MySQL/MariaDB and phpMyAdmin on the researchers' own computers for developing and testing the backend before deployment.

H2 Database. This is the in-memory database used only while running the backend's automated tests, so testing never affects the real school data.

Apache PDFBox. This is the Java library used to generate the printable bubble answer sheets and the downloadable PDF reports, such as the class record and item analysis.

Apache POI. This is the Java library used to generate the downloadable Excel (.xlsx) versions of the system's reports.

Google ZXing (Zebra Crossing). This is the QR code library used to generate the unique QR code printed on every answer sheet, which the mobile app scans to identify the exact test and student it belongs to, as well as the QR code used to set up two-factor authentication.

Jackson. This is the library, bundled with Spring Web, used to convert the backend's data to and from JSON whenever it exchanges information with the mobile app or the web dashboard.

Brevo. This is the third-party email delivery service connected to Spring Boot Mail so that verification and registration emails actually reach a teacher's real inbox.

Maven. This is the build tool used to manage the backend's external libraries and to compile, package, and run the application.

Postman. This is the tool used to manually test the backend's REST API endpoints during development, before the mobile app and web dashboard were connected to them.

Git and GitHub. These are the version control tool and hosting platform used to track source code history and to share and combine backend, mobile, and web dashboard code among the research team.